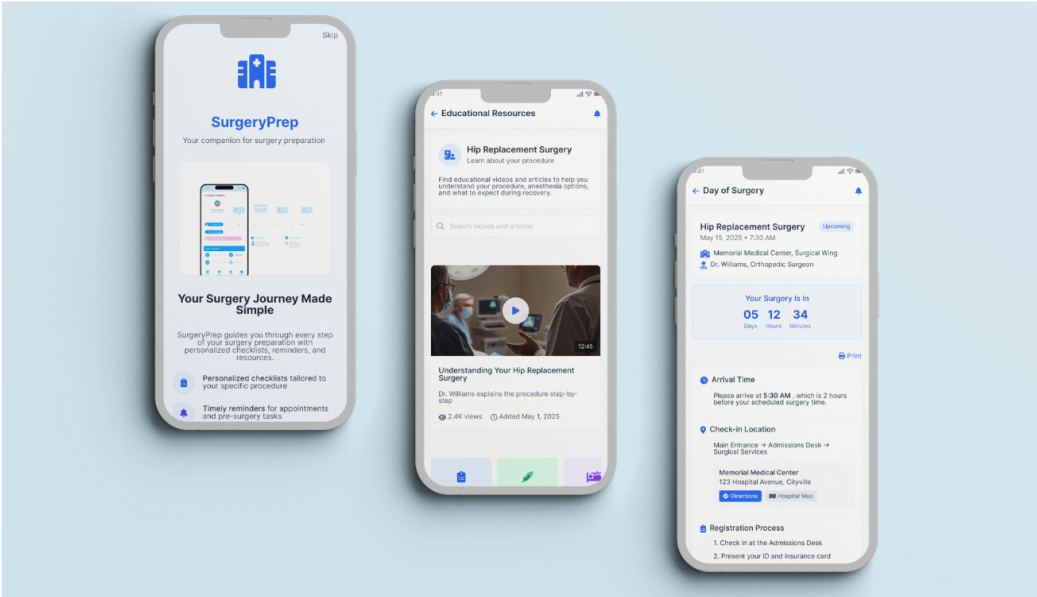




SurgeryPrep Mobile App Design

Role

UI/UX Designer

Duration

3 months

Team

Solo Designer

The Challenge

"I felt completely lost after my surgery consultation. The leaflet they gave me was confusing, and I couldn't remember half of what the doctor said. I ended up calling the hospital three times just to ask basic questions."

— Sarah, 34, Knee surgery patient

Patients undergoing surgery face a complex journey filled with anxiety, confusion, and information overload. Through analysis, it became clear that traditional communication methods—verbal instructions and printed leaflets—often fail to provide the right information at the right time, leading to frustrated patients and overwhelmed healthcare staff.

The Problem Identified

- Patients forget 40-80% of medical information immediately
- Printed materials get lost or misunderstood
- Healthcare staff spend hours answering repetitive questions
- Patient anxiety peaks due to lack of clear guidance

Solution Approach

SurgeryPrep was designed as a personalised mobile companion that delivers the right information at the right time, empowering patients throughout their surgical journey while reducing the communication burden on healthcare professionals.

Design Process



1 Research: Understanding the Human Experience

What needed to be discovered: Understanding the emotional and practical challenges patients face throughout their surgical journey, from pre-op anxiety to post-surgery recovery confusion.

Through systematic research, the goal was to identify pain points that could be addressed through thoughtful design, while also understanding healthcare providers' communication challenges.

"We answer the same five questions about post-surgery care at least 20 times per day. If patients had access to this information when they needed it, we could focus more on complex cases."

— Dr. Martinez, Orthopedic Surgeon

Research Approach

I conducted primary research to understand both patient and healthcare provider perspectives, combining qualitative interviews with journey mapping to capture the full surgical experience.

Patient Interviews: 8 participants were interviewed across various surgery types (orthopedic, cardiac, general) to understand diverse experiences and needs.

Healthcare Professional Interviews: Conversations with 5 clinicians (3 surgeons, 2 nurses) revealed their communication challenges.

Journey Mapping: Emotional highs and lows throughout the surgical process were documented to identify intervention opportunities.

Pain Point Analysis: Critical gaps in communication and support throughout the patient journey were systematically identified.

Research Synthesis

This section shares key insights from patient and healthcare professional interviews. Their perspectives helped uncover challenges and needs in the surgical journey, guiding the design of the Surgery Prep App.

Patient Interview Report

Patient journey: Pre-Surgery - Anxiety about upcoming surgery, need for information and reassurance	Patient journey: Day Before Surgery - Nervousness, need for clear instructions and emotional support	Patient journey: Surgery Day - Stress and fear, need for comfort and understanding of the process	Patient journey: Post-Surgery - Relief and recovery, need for guidance on aftercare and emotional support
Practitioner journey: Pre-Surgery - Preparation and planning, need for effective communication with patient	Practitioner journey: Day Before Surgery - Final checks and patient reassurance, need for clear communication	Practitioner journey: Surgery Day - Focus and precision, need for patient comfort and clear explanations	Practitioner journey: Post-Surgery - Monitoring and support, need for effective communication on recovery process
Key Communication Gap: Lack of clear instructions and emotional support before surgery	Key Communication Gap: Need for better emotional support and reassurance during surgery	Key Communication Gap: Insufficient guidance and emotional support after surgery	

Healthcare Professional Interview Report

Patient Anxiety: Patients often feel stressed and confused before surgery, impacting their overall experience and recovery.	Communication Gaps: Healthcare providers rely on verbal instructions and postures but struggle in addressing patient concerns effectively.	Continuous Training: Providing regular training for staff to keep up with the latest healthcare practices.	Innovation Integration: Incorporating newer technologies and practices to improve healthcare delivery.
Information Overload: Patients receive too much information at once, making it difficult to retain important details.	Staff Challenges: Healthcare staff feel overwhelmed by the need to repeatedly clarify information, affecting their efficiency.	Advanced Communication: Using digital platforms to enhance patient-staff communication.	Patient-Centric Design: Designing healthcare experiences that prioritize patient needs and preferences.
Time Constraints: Limited time to spend with each patient, making it challenging to address all concerns thoroughly.	Emotional Toll: Constantly managing patient expectations and emotions can be emotionally exhausting for healthcare workers.	Process Optimization: Streamlining healthcare processes to increase patient flow and reduce wait times.	Data Utilization: Leveraging patient data to personalize care and improve outcomes.

Critical Finding #1

Patients experience peak anxiety 24-48 hours before surgery due to uncertainty about preparation steps.

Critical Finding #2

Post-surgery recovery instructions are poorly retained, leading to repeated calls to healthcare providers.

Critical Finding #3

Personalised, timeline-based information delivery significantly improves patient confidence and adherence.

2 Insights: Defining the Design Challenge

What I needed to solve: I synthesised research findings to identify core user needs and define specific design challenges that would guide solution development.

My focus centered on transforming complex medical information into actionable, timely guidance that would reduce anxiety and improve patient outcomes.

Design Questions I Framed

- How might we help patients feel confident and prepared throughout their surgical journey?
- How might we reduce repetitive communication burden on healthcare staff?
- How might we deliver the right information at exactly the right time?
- How might we transform anxiety into empowerment through better design?

User Personas

I created detailed personas based on my research to maintain user-centered focus throughout the design process. These personas guided every design decision and feature prioritisation I made. I developed comprehensive character profiles that captured not just demographics, but emotional states, motivations, and specific pain points that emerged during interviews.

Primary Patient Persona

Julie Brooks
Profession: Molecular Biologist at a research institute near London. Her work is intellectually demanding, requiring intense focus and precision.
Family: Married. They have three children: Max (15), Lily (12), and Charlie (8).
Current Situation: Preparing for an ACL reconstruction surgery on her knee after having a long and tiring day at work. She is worried about missing a step of the final, crucial recovery process.

Concerns About During the Surgery:
The Unknowns of Anesthesia: The idea of general anesthesia is deeply unsettling. She's worried about the loss of control.
Vulnerability: The thought of being unconscious, completely reliant on the surgical team, is terrifying for someone used to being independent and in charge.
The Physical Ask: While she understands the procedure, the sheer scale of surgical instruments inside her body cutting and reshaping makes her nervous. She worries about missing a step of the final, crucial recovery process.

Concerns About After the Surgery:
Immediate Pain: This worried about managing it, knowing that unbearable pain on the post-operative day could have a side effect of being immobile.
Recovery Timeline & Uncertainty: How long will it take to walk without crutches? How long will it take to get back to work?
Adaptation: She anticipates physical therapy will be painful and demanding.
Complications: Post-surgical complications like infection, blood clots, or stiffness. She feels herself mentally running through

PERSONALITY
Introverted — Extroverted
Analytical — Creative
Risk — Time rich
Money — Organized
Independent — Team player

INFLUENCERS
What products or services are influencing her?
Facebook, LinkedIn, Instagram, YouTube, Google, Amazon, Apple, Microsoft, Tesla, SpaceX, Netflix, Disney, Netflix, Disney, Netflix, Disney

BEHAVIOR TRAITS
PASSIVE — ASSERTIVE
CALM — ANXIOUS
CONFIDENT — INSECURE

OVERALL MINDSET
Julie has a strong and out of her depth. She's trying to approach it logically, researching the procedure and recovery, but the emotional weight of the unknowns, the physical ask, and the significant disruption to her carefully balanced life are causing considerable stress. She wants to move with, but she's the process and the challenging recovery period ahead.

INTERESTS
Family Time: She's the busy schedule, the shared family activities, game nights, bedtime with her kids, and her partner.
Science & Learning: She's deeply passionate about her field of molecular biology and a recent acquisition with a new laboratory (including journals, attending what seminars).
Active Lifestyle: She enjoys physical activities like running and cycling with her family, contributing to her frustration with the current injury.

Healthcare Provider Persona

John Martin
Profession: Dr. Martin is a highly experienced orthopedic surgeon known for his precision in surgery and strong patient outcomes. He runs a tight schedule and an efficient and steady stream of pre-surgical visits, he has been selected by the volume of high-quality patients from patients.
Family: Single.
Current Situation: He's a busy of 4 and 5 above, making sure patients during and after surgery.

Concerns About During the Surgery:
Immediate Pain: This worried about managing it, knowing that unbearable pain on the post-operative day could have a side effect of being immobile.
Recovery Timeline & Uncertainty: How long will it take to walk without crutches? How long will it take to get back to work?
Adaptation: She anticipates physical therapy will be painful and demanding.
Complications: Post-surgical complications like infection, blood clots, or stiffness. She feels herself mentally running through

PERSONALITY
Introverted — Extroverted
Analytical — Creative
Risk — Time rich
Money — Organized
Independent — Team player

INFLUENCERS
What products or services are influencing her?
Facebook, LinkedIn, Instagram, YouTube, Google, Amazon, Apple, Microsoft, Tesla, SpaceX, Netflix, Disney, Netflix, Disney, Netflix, Disney

BEHAVIOR TRAITS
PASSIVE — ASSERTIVE
CALM — ANXIOUS
CONFIDENT — INSECURE

OVERALL MINDSET
Julie has a strong and out of her depth. She's trying to approach it logically, researching the procedure and recovery, but the emotional weight of the unknowns, the physical ask, and the significant disruption to her carefully balanced life are causing considerable stress. She wants to move with, but she's the process and the challenging recovery period ahead.

INTERESTS
Family Time: She's the busy schedule, the shared family activities, game nights, bedtime with her kids, and her partner.
Science & Learning: She's deeply passionate about her field of molecular biology and a recent acquisition with a new laboratory (including journals, attending what seminars).
Active Lifestyle: She enjoys physical activities like running and cycling with her family, contributing to her frustration with the current injury.

3 Ideation: From Problems to Solutions

Solutions I explored: I used structured ideation techniques to generate diverse concepts, then systematically evaluated them and prioritised based on user impact and technical feasibility.

The goal was to create solutions that would seamlessly integrate into patients' lives while addressing the core pain points identified through research.

Concept Exploration Process

I created sketches using Crazy 8s and other brainstorming techniques, generating over 20 concepts. A value vs. effort matrix was then created to prioritise the most impactful ideas that centred around creating a personal surgical assistant.

Concept Sketches

The ideation process was documented to show concept evolution and decision rationale.

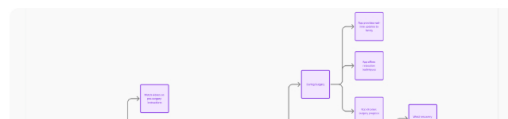


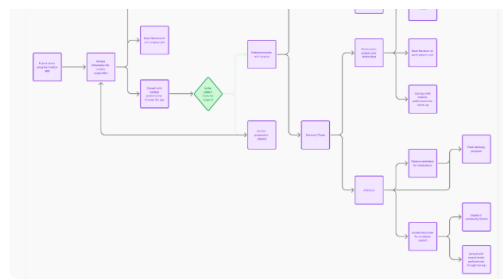
User Journey

I mapped out critical user journeys to understand how patients would navigate from initial diagnosis through complete recovery. Key touchpoints were identified where the design could provide maximum value and support.

User Journey Map

Comprehensive flow documentation was created to guide development priorities.

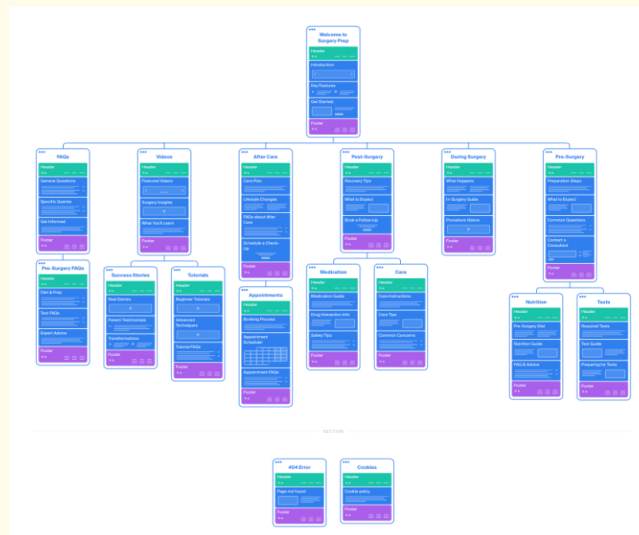




Based on research insights and user needs, six essential features were identified that would form the foundation of the solution.

- **Recovery Tracking**
Symptom logging with intelligent alerts for concerns
- **Gentle Reminders**
Timely notifications without overwhelming users
- **Educational Content**
Bite-sized, relevant information delivered progressively

I created the information architecture to organise the main features that I prioritised in the previous sections, ensuring intuitive navigation throughout the patient journey.



How the solution came to life: Concepts evolved through iterative design and testing, creating a comprehensive design system that prioritises accessibility and emotional comfort while maintaining medical credibility.

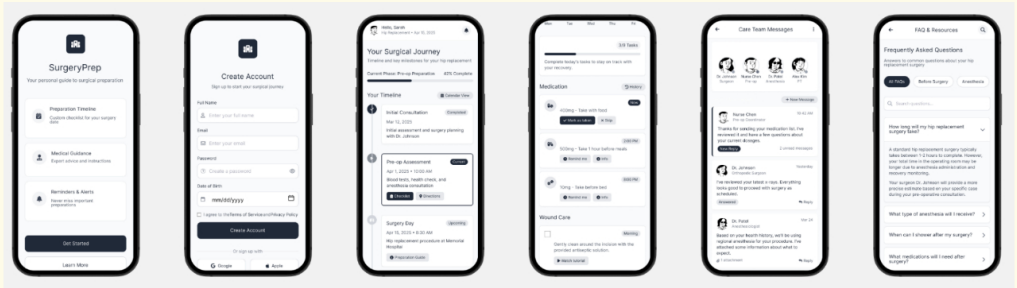
Design Evolution Process

The process began with low-fidelity wireframes to establish the core user flow and information hierarchy, then

systematically evolved these into a comprehensive design system. Every design decision was guided by research insights and user feedback.

Low-Fidelity Wireframes

I created initial wireframes to test core concepts and user flows before moving on to high-fidelity design work.

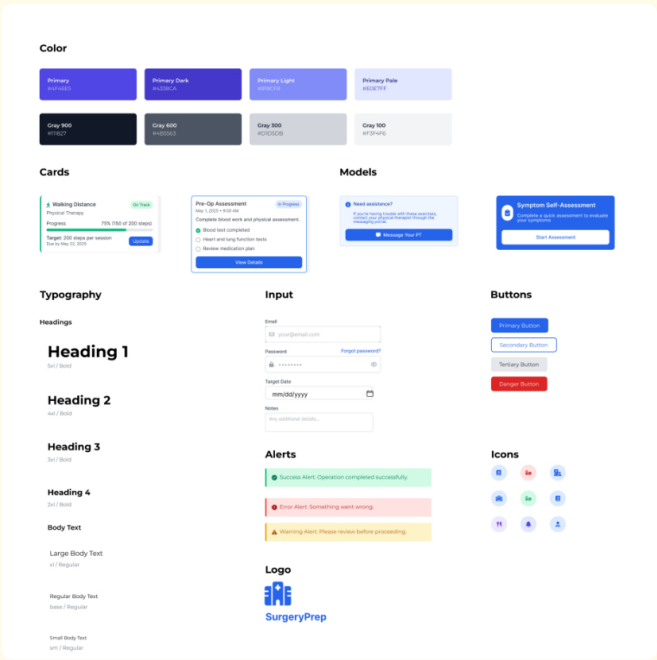


Design System Development

Next I worked on the design system which prioritised a calm, reassuring aesthetic while maintaining medical credibility. Every color, typography choice, and interaction was designed to reduce anxiety and build confidence based on psychological design principles.

Design System

A scalable design system was established to ensure consistency and efficiency across all touchpoints.

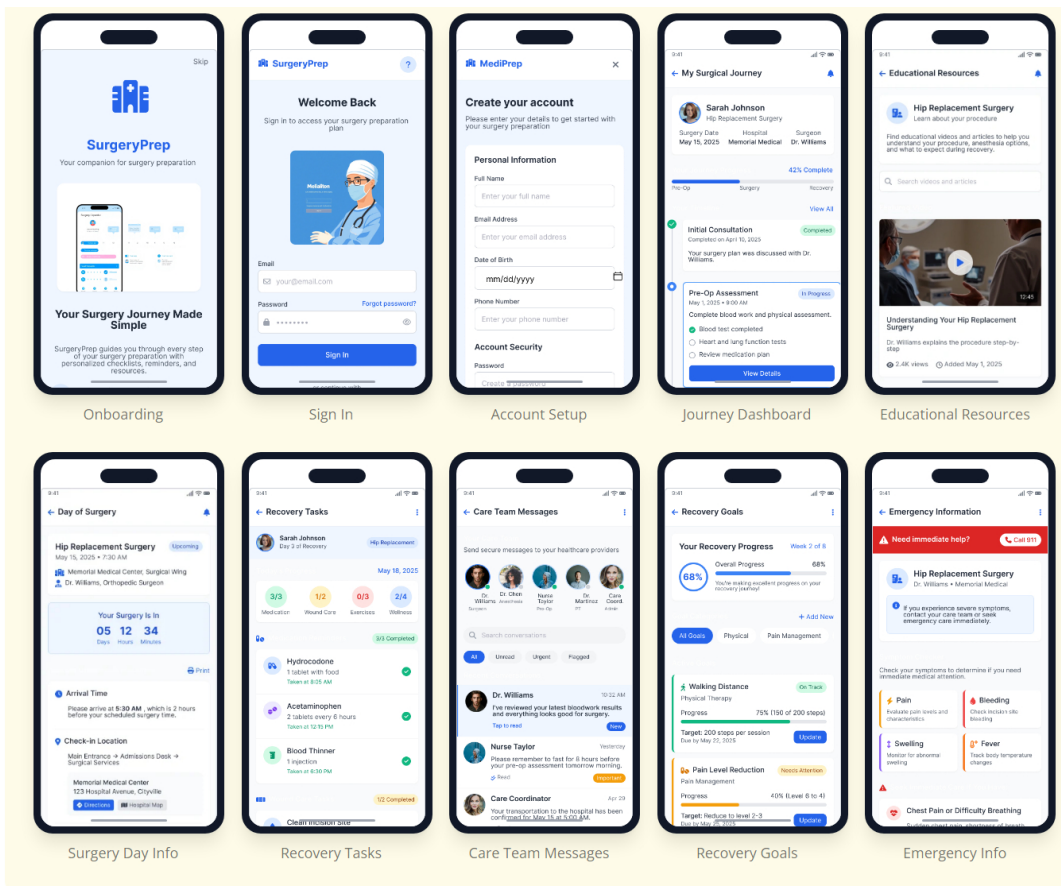


Final Design & User Validation

I conducted usability testing with five users (three patients, two clinicians) through task-based scenarios. All core user flows were tested and iterated based on feedback to ensure intuitive navigation and clear information hierarchy.

High-Fidelity Prototype

I moved on to create a fully interactive prototype demonstrating all key features and user flows for validation and stakeholder buy-in. The prototype incorporated all design system elements and provided a realistic preview of the final application.



What Was Validated

- 100% task completion rate across all users
- "Reassuring" and "much clearer than leaflets" feedback
- Clinicians confirmed significant call reduction potential
- Intuitive navigation requiring no explanation

Improvements Made

- Simplified medical terminology based on user feedback
- Added voice-over capability for accessibility
- Enhanced visual hierarchy for key information
- Refined notification timing to avoid overwhelm

5 Results: Impact & Next Steps

What was achieved: A solution was successfully created that addresses core patient pain points while providing measurable benefits to healthcare providers, validating the design approach through user testing and stakeholder feedback.

The project demonstrates how thoughtful UX design can transform healthcare experiences by bridging the gap between medical expertise and patient empowerment.

Key Learnings from This Project

Timing is Everything

The right information at the wrong time creates more anxiety, not less. Context-aware delivery is crucial for healthcare UX.

Language Matters

Medical jargon creates barriers. Simple, empathetic language builds trust and understanding.

Personalisation Scales

Surgery-specific content dramatically improves relevance without exponentially increasing complexity.

Next Steps Recommended

- **Pilot Program**
Partner with a local hospital for 3-month pilot study to validate real-world impact
- **Enhanced Accessibility**
Implement full voice support and multilingual capabilities
- **EHR Integration**
Explore integration with existing hospital record systems
- **AI-Powered Insights**
Develop predictive features for personalised care

Final Reflection

Through SurgeryPrep, thoughtful design was demonstrated to transform healthcare experiences. By focusing on human needs rather than institutional convenience, more than an app was created—a bridge between medical expertise and patient empowerment was built. The greatest validation came not from usability metrics, but from a patient who said: *"For the first time, I felt like I knew what was happening to my body."* This project reinforced the belief that great UX design has the power to reduce anxiety, build confidence, and ultimately improve lives.

